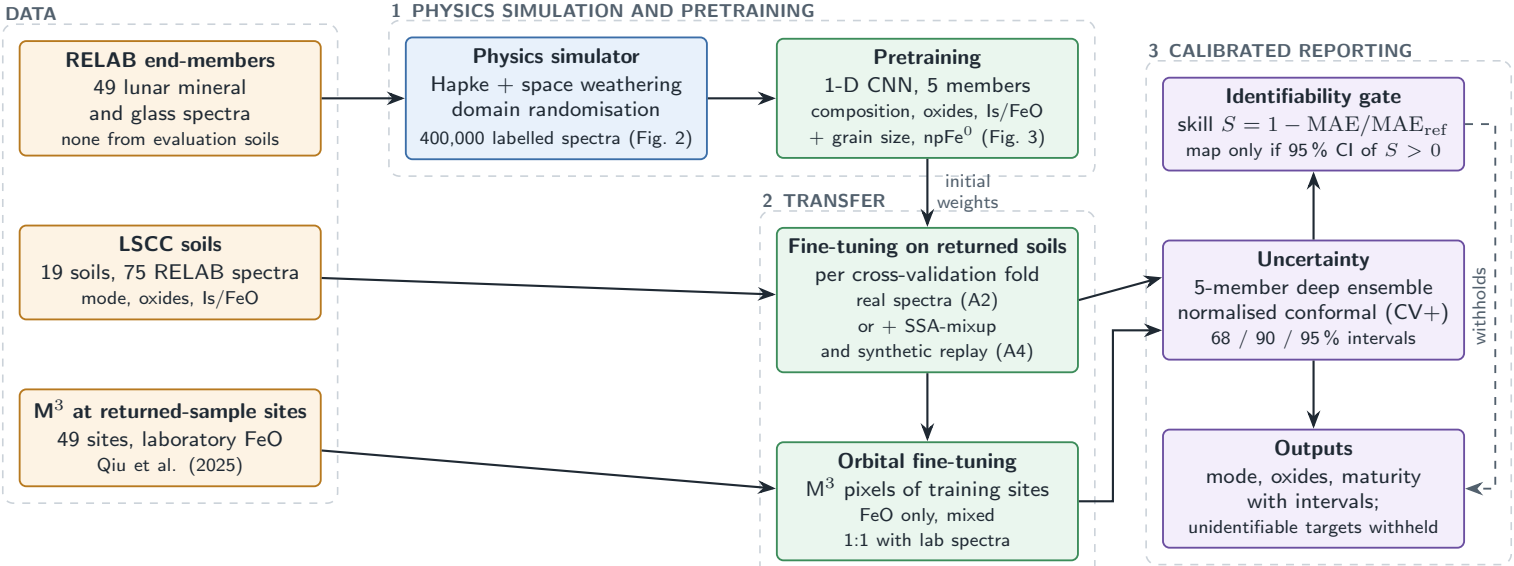

DATA
RELAB end-members
49 lunar mineral and glass spectra
none from evaluation soils
LSCC soils
19 soils, 75 RELAB spectra
mode, oxides, Is/FeO
M3 at returned-sample sites
49 sites, laboratory FeO
Qiu et al. (2025)
1 PHYSICS SIMULATION AND PRETRAINING
Physics simulator
Hapke + space weathering domain randomisation
400,000 labelled spectra (Fig. 2)
Pretraining
1-D CNN, 5 members
composition, oxides, Is/FeO + grain size, npFe0 (Fig. 3)
initial weights
2 TRANSFER
Fine-tuning on returned soils
per cross-validation fold
real spectra (A2)
or + SSA-mixup and synthetic replay (A4)
Orbital fine-tuning
M3 pixels of training sites
FeO only, mixed 1:1 with lab spectra
3 CALIBRATED REPORTING
Identifiability gate
skill $S = 1 - \mathrm{MAE}/\mathrm{MAE}_{\mathrm{ref}}$
map only if 95 % CI of $S > 0$
Uncertainty
5-member deep ensemble
normalised conformal (CV+)
68 / 90 / 95 % intervals
Outputs
mode, oxides, maturity with intervals;
unidentifiable targets withheld
withholds